

```

Using device: cpu
Using single-amplitude feyn data from /var/folders/1s/69py7bdx0g7g0t4_8rjh6g9h0000gn/T/single_amplitude_eval_hx4tsy6q/first_amplitude.csv
Imported 1 token rows
Wrote raw expression data to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20.csv
Wrote normalized token data to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_tok.csv
Input token limit: unlimited; longest input: 899 content / 901 with BOS/EOS
Numerical evaluator: gravity (processes={'3s2h': 1}; reference_modes=first,last,random; gauge_shift=True; tol_abs=2e-09; tol_rel=2e-08)
Loading model from /Users/kymani/Documents/Big_YM_Data_Sets/Cyclic_Gravity/Model/best_model.pt

Running decode mode: beam; max_length=256; beam_size=20; mask_invalid_tokens=False
  wrote human CSV to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_beam_human.csv

[beam] n=1
  top-1 correct    :    0 / 1  (0.0%)
  any-beam correct:    0 / 1  (0.0%)
  avg valid beams : 20.0
  shorter equivalent outputs: 0 / 1
  copied inputs: 0 / 1
  independent target: unavailable; exact-target metrics are not applicable

Example predictions (1 rows):
  row 0 [beam]
    target simple :
    input scram   : ((e_4 · e_5)*(e_4 · e_5))/6 + ((e_4 · p_5)*(e_4 · p_5)*(e_5 · p_4)*(e_5 · p_4))/(6*(p_4 · p_5)*(p_4 · p_5)) - ((e_4 · p_2)*(e_4 · p_2)*(e_5 · p_3)*(e_5 · p_3))/((p_2 · p_4)*(p_3 · p_5*p_4*p_5*p_4*p_5))
    top1 pred      : (p_2·F_4·p_5*p_1·F_4·p_5*p_1·F_5·p_4*p_1·F_5·p_4)/(p_1·p_4*p_1·p_5*p_2·p_3*p_4·p_5*p_4·p_5*p_4·p_5)
    top1 metrics   : token=None string=None num_simple=None num_scr=0
    any-beam       : token=None string=None num_simple=None num_scr=0
  wrote detail results to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_beam_results.csv

Wrote summary to /Users/kymani/Documents/New_Scattering_AI/Repo/gravity-evaluator-improvements/after/user_beam20_summary.csv
Total wall time: 9.86s

Terminal output PDF: /Users/kymani/Documents/New_Scattering_AI/Repo/Amplitudes-simplified-ML/data_testing/outputs/Gravity_5_Evaluation_Information_2026-09-23_01-19-10-132545.pdf

```